Assignment – 4

Student Name: Yashi Shakya
Enrollment Number: 23301012024
College Name: IGDTUW

Part A: Dataset Understanding

Q1. Load the dataset and display the first five records.

```
# Load Dataset and Display First Five Records
df = pd.read_csv('Dataset 2.csv')

print(df.head())
```

```
   UserID  Age  Gender SubscriptionType  WatchHoursPerWeek  DevicesUsed  \
0    1001   22  Female            Basic                 23            5
1    1002   55    Male            Basic                  9            4
2    1003   49    Male            Basic                  8            3
3    1004   39  Female          Premium                 19            5
4    1005   38  Female          Premium                 23            5

  FavoriteGenre  AdClicks  MonthlySpend SubscriptionRenewed
0        Comedy        13           353                  No
1         Drama        14           317                 Yes
2        Comedy        16           309                  No
3         Drama        45           833                 Yes
4        Sci-Fi        24           804                 Yes
```

Q2. Determine the number of rows and columns in the dataset.

```
print("rows and columns of dataset:", df.shape)
```

```
rows and columns of dataset: (750, 10)
```

Q3. Display all column names.Identify numerical and categorical features.

```
print(df.columns)
```

```
Index(['UserID', 'Age', 'Gender', 'SubscriptionType', 'WatchHoursPerWeek',
       'DevicesUsed', 'FavoriteGenre', 'AdClicks', 'MonthlySpend',
       'SubscriptionRenewed'],
      dtype='object')
```

Q4. Identify numerical and categorical features.

```
print("Numerical Features:")
print(df.select_dtypes(include=['int64','float64']).columns)

print("\nCategorical Features:")
print(df.select_dtypes(include=['object']).columns)
```

```
Numerical Features:
Index(['UserID', 'Age', 'WatchHoursPerWeek', 'DevicesUsed', 'AdClicks',
       'MonthlySpend'],
      dtype='object')

Categorical Features:
Index(['Gender', 'SubscriptionType', 'FavoriteGenre', 'SubscriptionRenewed'], dtype='object')
```

Q5. Check whether the dataset contains missing values.

```
print(df.isnull().sum())
```

```
UserID                 0
Age                    0
Gender                 0
SubscriptionType       0
WatchHoursPerWeek      0
DevicesUsed            0
FavoriteGenre          0
AdClicks               0
MonthlySpend           0
SubscriptionRenewed    0
dtype: int64
```

Part B: Exploratory Data Analysis

Q6. Calculate the average age of users.

```
print("Average Age =", df['Age'].mean())
```

```
Average Age = 41.824
```

Q7. Determine the average watch hours per week.

```
print("Average Watch Hours =", df['WatchHoursPerWeek'].mean())
```

```
Average Watch Hours = 14.236
```

Q8. Find the average monthly spending of users.

```
print("Average Monthly Spending =", df['MonthlySpend'].mean())
```

```
Average Monthly Spending = 689.9053333333334
```

Q9. Count the number of users in each subscription category.

```
print(df['SubscriptionType'].value_counts())
```

```
SubscriptionType
Basic      342
Premium    279
VIP        129
Name: count, dtype: int64
```

Q10. Determine the percentage of users who renewed their subscriptions.

```
renew_percent = (df['SubscriptionRenewed'].value_counts(normalize=True))*100

print(renew_percent)
```

```
SubscriptionRenewed
No     53.733333
Yes    46.266667
Name: proportion, dtype: float64
```

Part C: Data Preparation

Q11. Convert categorical features into numerical form.

```
encoder = LabelEncoder()

df['Gender'] = encoder.fit_transform(df['Gender'])

df['SubscriptionType'] = encoder.fit_transform(df['SubscriptionType'])

df['FavoriteGenre'] = encoder.fit_transform(df['FavoriteGenre'])

df['SubscriptionRenewed'] = encoder.fit_transform(df['SubscriptionRenewed'])

print(df.head())
```

```
...    UserID  Age  Gender  SubscriptionType  WatchHoursPerWeek  DevicesUsed  \
0    1001   22       0                 0                 23            5
1    1002   55       1                 0                  9            4
2    1003   49       1                 0                  8            3
3    1004   39       0                 1                 19            5
4    1005   38       0                 1                 23            5

   FavoriteGenre  AdClicks  MonthlySpend  SubscriptionRenewed
0              1        13           353                    0
1              2        14           317                    1
2              1        16           309                    0
3              2        45           833                    1
4              5        24           804                    1
```

Q12. Define the feature set (X) and target variable (y) for subscription renewal prediction.

```
X = df.drop(['SubscriptionRenewed'], axis=1)

y = df['SubscriptionRenewed']
```

Q13. Split the dataset into training and testing sets.

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print(X_train.shape)
print(X_test.shape)
```

```
... (600, 9)
    (150, 9)
```

Part D: Decision Tree Classification

Q14. Train a Decision Tree model to predict whether a user will renew their subscription.

```
dt_model = DecisionTreeClassifier(random_state=42)

dt_model.fit(X_train, y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

Q15. Evaluate the model using accuracy.

```
dt_pred = dt_model.predict(X_test)

dt_accuracy = accuracy_score(y_test, dt_pred)

print("Decision Tree Accuracy =", dt_accuracy)
```

```
Decision Tree Accuracy = 0.5666666666666667
```

Q16. Generate and interpret the confusion matrix.

```
cm = confusion_matrix(y_test, dt_pred)

print(cm)
```

```
[[50 32]
 [33 35]]
```

Part E: K-Nearest Neighbors (KNN)

Q17. Train a KNN classifier with K = 5.

```
knn_model = KNeighborsClassifier(n_neighbors=5)

knn_model.fit(X_train, y_train)

knn_pred = knn_model.predict(X_test)
```

Q18. Compare the accuracy of KNN with the Decision Tree model.

```
knn_accuracy = accuracy_score(y_test, knn_pred)

print("Decision Tree Accuracy =", dt_accuracy)

print("KNN Accuracy =", knn_accuracy)
```

```
Decision Tree Accuracy = 0.5666666666666667
KNN Accuracy = 0.6
```

Part F: Linear Regression

Q19. Train a Linear Regression model to predict monthly spending.

```
X_reg = df.drop(['MonthlySpend'], axis=1)

y_reg = df['MonthlySpend']

X_train_reg, X_test_reg, y_train_reg, y_test_reg = train_test_split(
    X_reg,
    y_reg,
    test_size=0.2,
    random_state=42
)

lr_model = LinearRegression()

lr_model.fit(X_train_reg, y_train_reg)
```

```
▾ LinearRegression
LinearRegression()
```

Q20. Predict the monthly spending for a new user and interpret the result.

```
new_user = [[751,25,1,2,20,3,1,10,1]]

prediction = lr_model.predict(new_user)

print("Predicted Monthly Spend = ₹", prediction[0])
```

```
Predicted Monthly Spend = ₹ 1351.1233293485086
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid featu
  warnings.warn(
```

Business Reflection Answers

1. Which factors influence subscription renewal the most?

Age, subscription type, watch hours, favorite genre, monthly spending, and ad clicks are likely to influence renewal.

2. Why is subscription renewal a classification problem?

Because the output has two categories: Yes or No.

3. Why is monthly spending a regression problem?

Because monthly spending is a continuous numerical value.

4. Which algorithm performed better?

Compare the accuracies of Decision Tree and KNN. The model with higher accuracy is better.

5. How can Netflix improve customer retention?

Netflix can identify users likely to leave and provide personalized recommendations, discounts, and premium plans to retain customers.